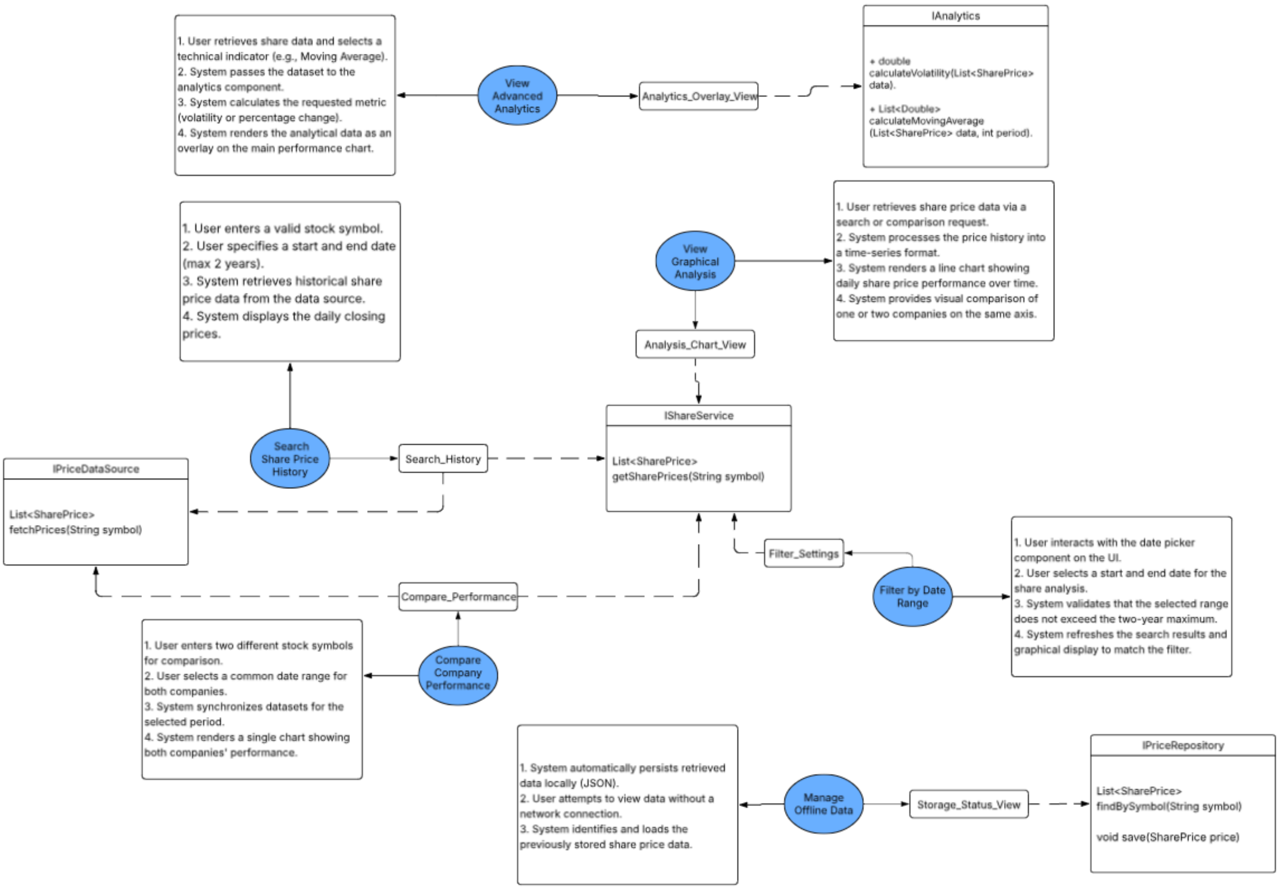
System Interface



Data Retrieval and Analysis

For the 'Search Share Price History', users enter a symbol and a date range of up to two years. The 'Search_History' dialog triggers the 'IShareService' interface, specifically the 'getSharePrices(String symbol)' operation.

The 'Compare Company Performance' use case goal uses the Compare_Performance dialog to fetch and synchronize datasets for two different companies over a shared period. Like search, it relies on 'IShareService' to gather data.

Required Interface ('IPriceDataSource'): Both search and comparison rely on the system's ability to "obtain daily price information" from external sources like Yahoo Finance via the 'fetchPrices(String symbol)' operation.

Visual and Technical Processing

The 'View Graphical Analysis' facilitated by the 'Analysis_Chart_View', this use case renders retrieved data into line charts to show performance over time.

For the 'View Advanced Analytics', it is linked to the 'Analytics_Overlay_View', this interaction utilizes the 'IAnalytics' interface to perform "mathematical calculations". Operations include 'calculateVolatility' and 'calculateMovingAverage'.

The 'Filter by Date Range', through the 'Filter_Settings', the system ensures user input aligns with constraints, specifically enforcing the maximum two-year range for historical data retrieval.

Persistence and Offline Support

The 'Manage Offline Data' use case fulfills the requirement to "persistently store such price data" for functionality without a network connection.

For the interface (IPriceRepository), the 'Storage_Status_View' dialog interacts with this interface. It uses 'save(SharePrice price)' to store data in a persistent format (such as JSON) and 'findBySymbol(String symbol)' to retrieve cached data when the system is offline.